

The Effect of Contamination on Contamination Limit Regulation

US Coal Mining and Drinking Water Utilities

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Self-reported compliance

- Regulatory models requiring entities to self-report compliance:
 - ▶ Healthcare HIPAA violation disclosure; Energy firms self-report anti-competitiveness/negligence to the Federal Energy Regulatory Commission; Public companies' SOX self-report financial statement accuracy; Drinking water utilities' water quality.
- Incentive to shirk reporting. Consider drinking water utilities' federal water quality regulation:
 - ▶ No self-report $\Rightarrow$ no contamination exceedance violation issued (Naylor, 2024; Zou, 2021).
 - ▶ No self-report saves self-reporting costs (Oleckno, 1982).
- **Incentives rise with contamination**
 - ▶ $\uparrow$ contamination $\rightarrow \uparrow$ testing requirements ($\uparrow$ **reporting costs**)
 - ▶ $\uparrow$ contamination $\rightarrow \uparrow$ probability contamination violation ($\uparrow$ **need to conceal**).
- Whether it is self-reporting costs or concealment has different policy implications for improving reporting.

Drinking water quality and coal mining

- Drinking water utilities provide treated water to 85% of US households.
 - ▶ Utilities source from groundwater and/or surface water, serve from twenty-five to millions of customers, publicly or privately owned.
- 9 to 34 million US households obtain water from a utility violating a health standard (Allaire, Wu, and Lall, 2018).
- **Coal mining is a source of water contamination**
 - ▶ **Acid mine drainage:** sulfuric acid formed from sulfide rock dissolves metal ions and sulfates, travels into surface and ground water (Skousen, Ziemkiewicz, and McDonald, 2019).
 - ▶ **Coal wash slurry:** impurities stored in retaining ponds seep through porous rock, or ponds fail.
- Federal policy is expanding mining access and delaying plant retirements (U.S. President (Trump) (2025); Daly (2025)).
- Coal supplied 16% of US electricity in 2025 (International Energy Agency (2025)).

Research questions

Does coal mining raise contamination at drinking water utilities?

What happens to firm self-reporting when contamination rises exogenously, under self-reported compliance?

Is any reduction in self-reporting concealment of exceedances, or avoidance of reporting costs?

What I do

- Setting: drinking water utilities downstream of coal mines, 1985–2005.
 - ▶ Contamination is an **upstream externality** — the firm cannot reduce it by cutting output (Bennear, 2005).
 - ▶ Separating contamination from utility choices helps identify the effect of contamination.
- Adapt Douglas and Wiggins (2015) instrument for coal production to address two sources of endogeneity:
 - ▶ Coal mines are near coal seams and transportation that independently predict water quality and utility characteristics.
 - ▶ Regulators block upstream mining for vulnerable utilities.
- Contributions:
 - ▶ Causal evidence that contamination reduces self-reporting.
 - ▶ Rationalize utility decisions to self-report in light of regulator's actions with a model.
 - ▶ First evidence that **the cost of self-reporting**, not only concealment, causes environmental under-reporting (Naylor, 2024; Zou, 2021).
 - ▶ First causal estimates of the effect of **coal mining on treated drinking water** at US utilities.

What I find

- ↑ mining → ↑ contaminant concentrations, but never above the MCL. 10M cumulative tons upstream raises mean nitrate by 11.0% of its sample mean (***) and selenium by 8.5% (**); arsenic and barium show no significant change.
- ↑ mining → ↑ **under-reporting**. One more upstream mine raises MR violations by 3–5 pp, against base rates of roughly 4–6%.
- ↑ mining → **no significant change** in MCL violations.
- ↑ mining → ↑ **inspection visits** (+2.1 pp).
- Formal enforcement: −3.4 pp with utility and year fixed effects, −1.1 pp (not significant) with state × year fixed effects, against a base rate of 3%.
- **Self-reporting costs, not concealment:**
 - ▶ When nitrate readings cross 50% of the MCL → monitoring requirements ↑ (reporting cost ↑) → self-reporting ↓.
 - ▶ Sanitary inspections at under-reporting utilities do not discover concealed violations.
 - ▶ Contaminant concentrations stay below MCL.

Contamination shifts monitoring from the firm onto the regulator.

Self-reporting model

Institutional setting

- Utilities federally regulated by the Safe Drinking Water Act (“SDWA”).
- Contaminants cannot exceed a Maximum Contaminant Level (“MCL”).
- Self-report to regulator or get a Monitoring and Reporting violation (“MR”).
- Self-reporting requirements depend on water contamination/threats to water.
- States are SDWA enforcers with discretion over enforcement and visits:
 - ▶ Sanitary survey - comprehensive on-site inspection
 - ▶ Technical assistance - helps fix problem or become compliant.
 - ▶ Sampling - prompted by suspected contamination event or violation.
 - ▶ Enforcement visit - triggered by suspected violation.
- Enforcement, while remedial rather than punitive, is still costly:
 - ▶ Informal (letters, notices, compliance plans)
 - ▶ Formal (consent decrees, fines, prosecution)
 - ▶ MR violations are punished less severely than MCL violations.

Model: the utility's decisions to self-report or under-report

Utility draws compliance cost type $\theta \sim F(\cdot; m)$, worsening in upstream contamination m .

- Choose negligence a , unobserved by regulator, with benefit $\theta b(a)$
- Opportunity cost relative to a maximum negligence $\theta b(\bar{a})$.

Utility draws a self-reporting cost $c \sim G(\cdot; m)$, worsening in upstream contamination m .

- A test reveals an MCL with probability $q(a)$, $q' \geq 0$.

Model: the utility's decisions to self-report or under-report

Penalties: r for a self-reported exceedance, t for an MR violation, $s > r$ if an inspection (probability p) uncovers a hidden exceedance.

$$\underbrace{c + q(a)r}_{\text{report}} \leq \underbrace{t + pq(a)s}_{\text{under-report}} \implies \text{report iff } c \leq \hat{c}(a) \equiv t - (r - ps)q(a).$$

- A system with **zero** exceedance risk still stops reporting once $c > t$.

$$e(a; c) = \min \{ c + q(a)r, \quad t + q(a)ps \}$$

- The utility chooses a to maximize $-\theta[b(\bar{a}) - b(a)] - e(a; c)$.

Model: why the utility under-reports

Proposition 2

A type- θ utility under-reports if and only if its cost draw exceeds this cutoff, $c > c^*(\theta)$, so the share of utilities under-reporting and receiving an MR violation is:

$$MR(m) = \int_0^{\bar{\theta}} [1 - G(c^*(\theta); m)] f(\theta; m) d\theta \equiv \int_0^{\bar{\theta}} h(\theta, m) f(\theta; m) d\theta.$$

Differentiating splits into two channels:

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m) d\theta}_{\text{Channel A: cost-distribution shift. Hold } \theta \text{ distribution fixed and change probability type } \theta \text{ utilities draw } c \text{ exceeding } c^*} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m} d\theta}_{\text{Channel B: type-distribution shift. Hold proportion of type } \theta \text{ utilities whose cutoff exceeds } c^* \text{ fixed and change the proportion of utilities of type } \theta.}$$

Data

- **Violations, enforcement, visits** — SDWIS/ECHO, universe of utilities 1985–2005. Binary by utility $\times$ year; days-in-year as a robustness measure.
 - ▶ Under-inclusive but accurate: 95% of the violations SDWIS holds are correctly reported (US EPA, 2000).
- **Contaminant concentrations** — EPA Six-Year Review round 2, 1998–2005. Cleaned following Ravalli et al. (2022); non-detects set to $MDL/\sqrt{2}$.
- Contaminants: **arsenic, nitrate, barium, selenium** — inorganic chemicals with stable utility MCLs 1985–2005.
- **Mines** — MSHA locations merged to EIA annual production, mine level.
- **Coal sulfur %** — USGS COALQUAL samples, averaged within 20 km of each watershed.
- **Hydrology** — NHDPlus HUC12 watershed boundaries and inflow/outflow.
- **Sample:** Utilities with a coal mine **within two watersheds upstream** of their intake and no coal mine colocated with their intake. 10,641 utility-years, 565 utilities.

Matching intakes to the watershed above them

- HUC12 is the finest watershed mapped for the continental US (>87,000 units).
- Exposure = mines and tons of coal in the HUC12s **within two watersheds upstream** of the utility's intakes.
- Average distance between centroid of adjacent HUC12s is 11 km.
- Intakes **colocated** with a mine are excluded.
- Multiple intakes: sum production, average sulfur.



MR violations are common; almost no MCL violations

Panel A: Any violation in year

Contaminant	MR		MCL	
	%	N	%	N
Nitrates	5.68	612	0.11	12
Arsenic	3.73	402	0.05	5
Inorganic chemicals	4.31	465	0.30	32

Panel B: Days in year

Contaminant	MR				MCL			
	Mean	SD	P90	P99	Mean	SD	P90	P99
Nitrates	14.54	65.35	0.00	365.00	0.27	9.19	0.00	0.00
Arsenic	9.98	54.32	0.00	365.00	0.08	3.83	0.00	0.00
Inorganic chemicals	11.53	58.47	0.00	365.00	0.71	15.08	0.00	0.00

Nitrate MR 5.7%, arsenic 3.7%, IOC 4.3% of utility-years. MCL violations: 12, 5, and 32 events across 1985–2005.

Mining exposure in the sample

Variable	Mean	SD	P90	P99
Number of mines upstream	1.12	2.27	4.00	11.00
Cumul. upstream coal prod. (10M ST) since 1985	0.32	1.00	0.68	5.23
Percentage of coal weight as sulfur	1.11	1.13	2.96	4.01

On average 1.1 mines upstream and 1.1% sulfur by weight; cumulative upstream production reaches 5.2 (10M ST) at the 99th percentile.

Does coal mining contaminate utility water?

Contaminant concentration summary statistics

Variable	MCL	Mean	Max	Std. Dev.	N	Near MCL
Panel A: Full sample						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0028	0.0110	0.0023	416	0.0%
Nitrate (mg/L)	10.0 mg/L	0.8020	5.5400	0.8738	488	0.4%
Barium (mg/L)	2.000 mg/L	0.0740	0.7071	0.0659	407	0.0%
Selenium (mg/L)	0.050 mg/L	0.0047	0.0354	0.0049	405	0.5%
Panel B: Utility exposed to above median cumulative tons of coal extraction						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0032	0.0100	0.0026	205	0.0%
Nitrate (mg/L)	10.0 mg/L	0.8254	5.5400	0.8748	260	0.4%
Barium (mg/L)	2.000 mg/L	0.0811	0.3320	0.0569	196	0.0%
Selenium (mg/L)	0.050 mg/L	0.0038	0.0354	0.0043	196	1.0%

Design: cumulative production, measured concentrations

$$Concen_{ct} = \beta \cdot \sum_{\tau=1985}^t \text{UpstreamCoalProd}_{c\tau} + \rho_c + HUC02_c \cdot \tau_t + \epsilon_{ct}$$

- $Concen_{ct}$: mean annual concentration at utility c , year t (SYR2, 1998–2005).
- β : effect of 10 million additional cumulative short tons mined upstream since 1985.
- ρ_c : utility fixed effects. $HUC02_c \cdot \tau_t$: HUC02-specific year effects, absorbing watershed-wide industrial trends.
- SEs clustered at the utility level.
- Continuous treatment, **no control**: identification comes from relative exposure.
- Assumption: utilities with large upstream production changes would have trended like those with small ones. (balance test on utility characteristics $\rightarrow$)

Mining raises contaminant levels

	Arsenic (1)	Nitrate (2)	Barium (3)	Selenium (4)
Panel A: Within one HUC12 upstream				
Cumul. upstream coal prod. (10M ST)	0.0022*** (0.0004)	0.0278 (0.1021)	0.0213** (0.0097)	0.0031** (0.0015)
Panel B: Within two HUC12 upstream				
Cumul. upstream coal prod. (10M ST)	0.0001 (0.0001)	0.0883*** (0.0257)	0.0136 (0.0112)	0.0004** (0.0002)
Observations	366	472	353	350

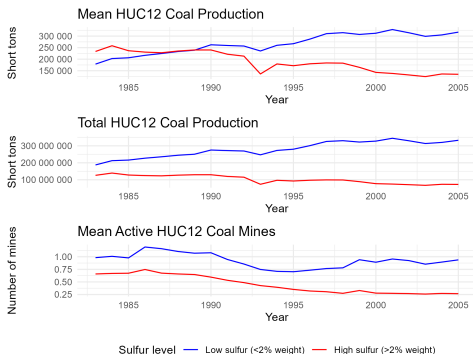
Notes: All specifications include utility and HUC02 $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Nitrate +0.09 mg/L per 10M tons (***): 11.0% of its mean. Selenium +0.0004 mg/L (**): 8.5% of its mean. **Every reading in the sample stays under its MCL.** (effect across choice of upstream watershed radius $\rightarrow$)

Do utilities keep self-reporting?

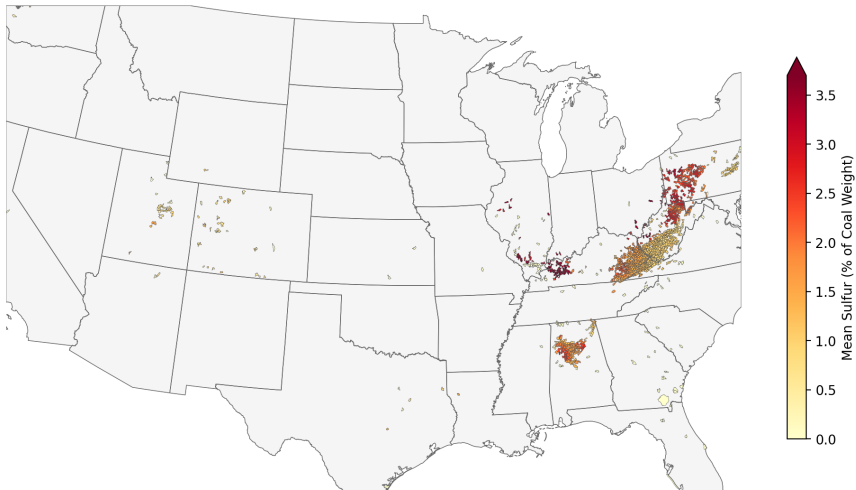
The Acid Rain Program (“ARP”) moved coal production for reasons unrelated to water quality

- Adapt Douglas and Wiggins (2015) ARP instrument for coal production to address endogeneity of mining and utility characteristics.
- 1990 Clean Air Act amendments cap SO_2 from power plants; Phase I deadline **1995**.
- Power plants were 90% of coal demand; switching to low sulfur coal was the main compliance response (Lattanzio, 2022).
- High sulfur mines have limited ability to wash the sulfur out (bound to chemical structure).

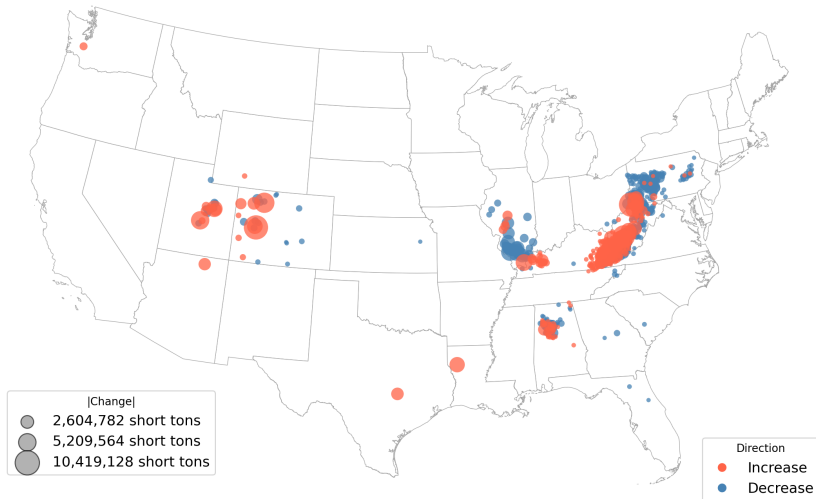


Watershed coal sulfur content

HUC12 watersheds containing coal mines, by mean coal sulfur content



Where coal production moved, 1985–2005



IV design: the ARP as a shifter of where coal is mined

First stage:

$$\text{NumMines}_{ct} = \alpha_0 + \alpha_1 \text{NumFacility}_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ + \text{State}_s \cdot Y_t + C_c + \nu_{ct}$$

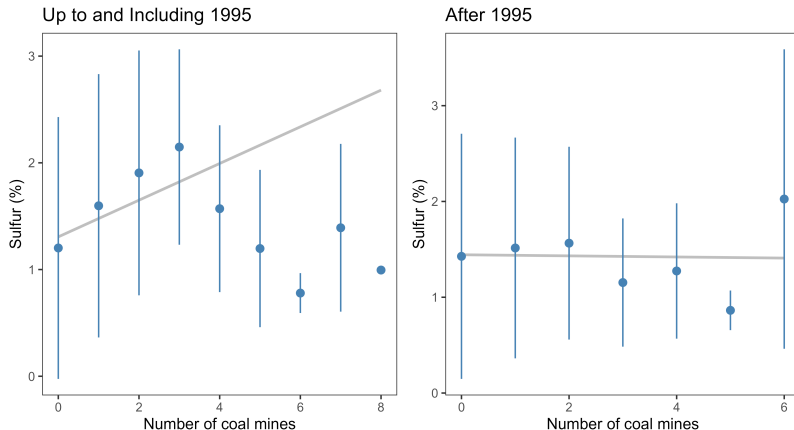
Second stage:

$$\text{Violation}_{vct} = \beta_0 + \beta_1 \text{NumFacility}_{ct} + \gamma \widehat{\text{NumMines}}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

- $\text{Post95}_t = \mathbf{1}\{t \geq 1995\}$. Sulfur is time-invariant, so C_c absorbs its level.
- Outcome: 1 if utility c had a violation of type v at any point in year t . Coefficients are in percentage points.
- SEs clustered at the utility level.

Shift in location of coal production w.r.t. sulfur

Descriptive evidence of first-stage relevance



Sample: HUC12 watersheds directly upstream of drinking water utility water intakes 1985-2005.

Points show the mean sulfur as a percent of coal weight within each coal-mine-count bin; ranges show ± 1 standard deviation within bin.

First stage: high sulfur watersheds lost their mines

	Upstream coal mines (sum) (1)
Post-1995 $\times$ Upstream sulfur %	-0.4933*** (0.0703)
F-test (1st stage, clustered)	49.20
Utility fixed effects	✓
State $\times$ year fixed effects	✓
Observations	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(first-stage strength across choice of upstream watershed radius $\rightarrow$)

Instrument validity

- **Independence.** The ARP is a national clean-air law set by national energy and emissions politics, not downstream utilities' water quality.
- **Exclusion.** Effect of instrumenting mining downstream of utility on utility behavior is not significant. (MR violations →), (visits →), (enforcement →). Utility characteristics are not predicted by instrument. (full balance table →)
- **Monotonicity.** A defier is a high-sulfur watershed that expanded *because* its coal was high sulfur. Scrubber adoption costs more likely lead to lost market share rather than gain; low-cost high-sulfur regions that expand anyway are never-takers.

Mining makes utilities stop reporting

	Monitoring and reporting (MR) violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.68** (0.33)	0.55 (0.34)	0.62** (0.31)	0.42 (0.33)	0.49 (0.32)	0.19 (0.34)
2SLS						
Upstream coal mines (sum)	5.12*** (1.70)	3.94** (1.75)	4.85*** (1.57)	3.78** (1.55)	4.66*** (1.62)	3.06* (1.66)
RF						
Post-1995 × Upstream sulfur %	-2.12*** (0.66)	-1.95** (0.83)	-2.01*** (0.60)	-1.87** (0.73)	-1.93*** (0.63)	-1.51* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** p<0.01, ** p<0.05, * p<0.1.

Base rates of nitrates, arsenic, and IOC are 5.7%, 3.7%, and 4.3%. OLS estimates are far smaller — consistent with negative omitted variable bias, as expected if poorly performing utilities (more MR violations) are protected from mine siting (fewer mines). (effect across choice of upstream watershed radius →)

Reporting cost, not concealment

- Nitrate rules require **more frequent monitoring** once a reading exceeds 50% of the MCL — a discrete jump in c , with no jump in the penalty for under-reporting and no MCL.
- Exceedance risk is held fixed (Channel B = 0), reporting cost increases (Channel A ≥ 0).

	Nitrate MR (1-yr) (1)	Nitrate MR (6-mon) (2)
Concen. > 50% MCL	58.9341** (24.5521)	26.0695** (10.5051)
Mean concen. (z-score)	1.2906 (1.4648)	0.0795 (0.9057)
Observations	835	835

Notes: All specifications include utility and year fixed effects. SEs clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Descriptive, not causal — read the sign, not the magnitude. Utilities under-report when self-reporting gets expensive, not to conceal exceedances.

How do regulators respond?

Enforcement does not counteract under-reporting

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	-0.15 (0.42)	-0.06 (0.44)	-1.00*** (0.21)	-0.12 (0.19)	0.33 (0.45)	-0.07 (0.48)
2SLS						
Upstream coal mines (sum)	-3.06 (2.15)	-0.71 (2.10)	-3.37*** (0.95)	-1.08 (0.86)	2.37 (2.11)	-0.42 (2.25)
RF						
Post-1995 × Upstream sulfur %	1.26 (0.84)	0.35 (1.03)	1.39*** (0.40)	0.53 (0.41)	-0.98 (0.85)	0.21 (1.11)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** p<0.01, ** p<0.05, * p<0.1.

One more upstream mine reduces formal enforcement by 3.4 pp with utility and year fixed effects alone (significant), but the effect loses significance once state × year fixed effects are added. (effect across choice of upstream watershed radius →)

The regulator increases inspections, not enforcement

	Sanitary (1)	Technical assistance (2)	Enforcement (3)	Sample collection (4)	Inspection (5)
OLS					
Upstream coal mines (sum)	0.28 (0.29)	0.10 (0.20)	0.04 (0.11)	-0.07 (0.05)	0.06 (0.13)
2SLS					
Upstream coal mines (sum)	0.78 (1.20)	0.67 (0.95)	1.07 (0.81)	0.48 (0.50)	2.13** (1.07)
RF					
Post-1995 × Upstream sulfur %	-0.39 (0.59)	-0.33 (0.47)	-0.53 (0.39)	-0.24 (0.24)	-1.05** (0.49)
Utility fixed effects	✓	✓	✓	✓	✓
Year fixed effects					
State × year fixed effects	✓	✓	✓	✓	✓
Utilities	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Inspection visits rise significantly (+2.1 pp). Regulators do not systematically step up the visit types tied to sanctions. Visits would reveal concealment and be associated with higher penalties. (effect across choice of upstream watershed radius →)

Conclusion

- Exogenous contamination raises the cost of self-reporting, and self-reporting falls — **even when utilities remain below MCL**.
- Under-reporting due to cost avoidance, rather than concealing a contamination violation.
- Regulators' inspection visits rise significantly — substituting for utility self-reporting.
- Enforcement response is mixed and fixed-effects-sensitive: formal action falls with baseline fixed effects but loses significance once state $\times$ year effects are added.
- **Policy.** Tightening MCLs does nothing if nobody tests. Raising self-reporting penalties could reduce under-reporting.
- Open question: whether regulator monitoring is more efficient than self-reporting.

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Appendix

Appendix: combined MR and MCL violations

	Nitrates		Any violation Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.67** (0.33)	0.54 (0.34)	0.60* (0.31)	0.43 (0.34)	0.56* (0.31)	0.28 (0.33)
2SLS						
Upstream coal mines (sum)	5.06*** (1.70)	3.89** (1.75)	4.94*** (1.57)	4.00** (1.56)	4.56*** (1.64)	3.09* (1.67)
RF						
Post-1995 × Upstream sulfur %	-2.09*** (0.66)	-1.92** (0.83)	-2.04*** (0.60)	-1.97*** (0.73)	-1.89*** (0.64)	-1.53* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The combined effect mirrors the MR effect: nitrate 3.9–5.1 pp, arsenic 4.0–4.9 pp, IOC 3.1–4.6 pp across the two fixed-effects specifications.

Appendix: instrument balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.9942*** (0.3834)
Post-1995 $\times$ Upstream sulfur	0.0509 (0.0386)	0.0563 (0.0376)
Utility fixed effects	✓	
Year fixed effects		✓
State $\times$ year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

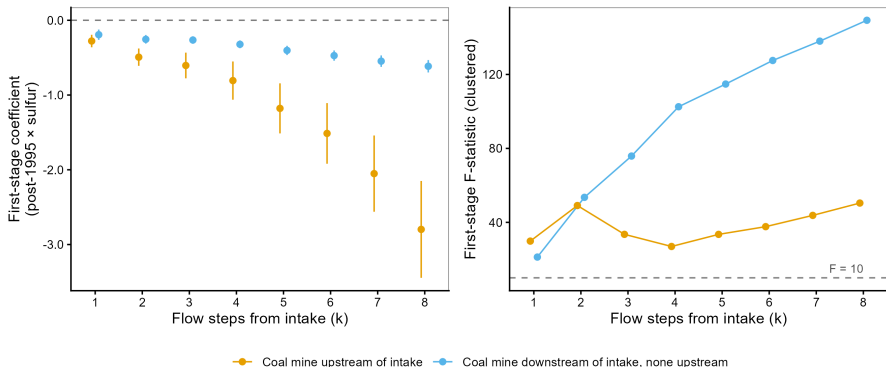
Appendix: balance test on utility characteristics

	Cumul. upstream coal prod. (10M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.2504 (0.2532)		-0.1596 (0.1198)	
Log backcast pop. 1997	0.1091 (0.0750)	0.1512 (0.0938)	0.0159 (0.0126)	0.0208 (0.0144)
Log pop. served (SYR2)	-0.0934 (0.0666)	-0.1113 (0.0956)	0.0060 (0.0259)	-0.0301 (0.0334)
Number of intake facilities	0.0089 (0.0569)	-0.0324 (0.0744)	0.0680* (0.0359)	0.0721* (0.0406)
Surface water source	0.1225 (0.1643)	-0.3902 (0.3296)	0.1210 (0.1307)	0.0521 (0.1601)
Public ownership	-0.1554 (0.2551)	-0.0599 (0.2240)	0.2084** (0.0911)	0.1048 (0.1098)
Wholesaler	-0.3780* (0.2041)	-0.7465 (0.9141)	-0.2164 (0.1340)	-0.2212 (0.1683)
Source water protection	-0.4549 (0.2749)	-0.4385 (0.4305)	-0.2807* (0.1521)	-0.1460 (0.1930)
Joint <i>F</i> -test (all covariates)	2.33	0.96	8.96	2.46
<i>p</i> -value	0.029	0.461	0.000	0.023
Observations	122	120	122	120
State fixed effects		✓		✓

Notes: Heteroskedasticity-robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

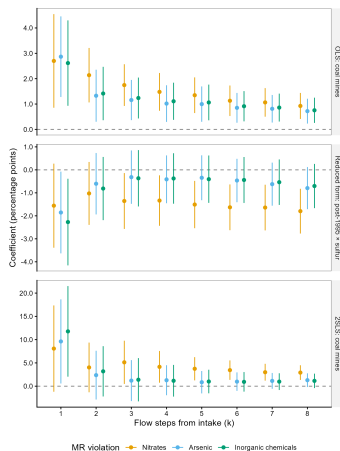
Observable characteristics jointly predict continuous upstream production only without state fixed effects ($p=0.029$) and lose significance once state fixed effects are added ($p=0.461$), as in the main specification. The joint test on the binary presence of upstream mining stays significant with or without state fixed effects.

Appendix: first stage across choice of upstream watershed radius



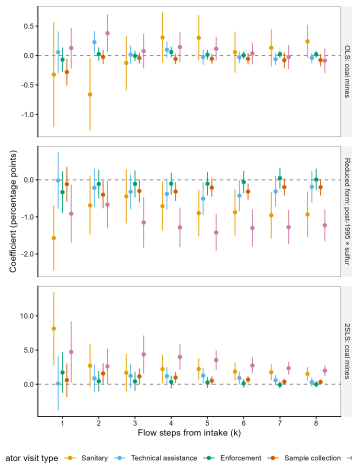
The first-stage coefficient and F-statistic for utilities downstream of a mine stay well above conventional weak-instrument thresholds as the upstream watershed radius k varies from one to eight flow steps.

Appendix: exclusion placebo, MR violations across k



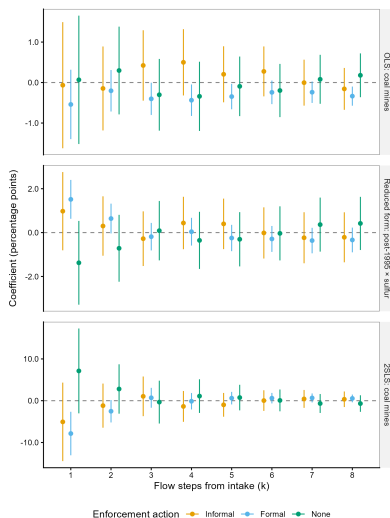
For utilities upstream of a mine — where mining cannot reach the intake hydrologically — OLS estimates on MR violations are positive and significant at every k . In the reduced form and 2SLS, nitrate estimates are significant for $k \geq 3$, while arsenic and inorganic chemical estimates are significant only at $k = 1$.

Appendix: exclusion placebo, visits across k



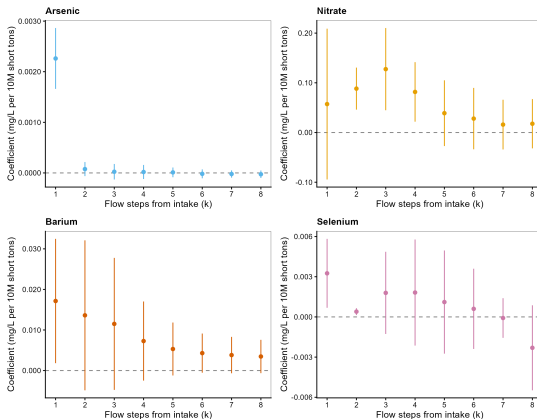
For utilities upstream of a mine, inspection visits rise significantly at nearly every k , sanitary visits at $k = 1$ and $k \geq 4$, and sample collection visits at several radii; technical assistance and enforcement visits are mostly insignificant.

Appendix: exclusion placebo, enforcement across k



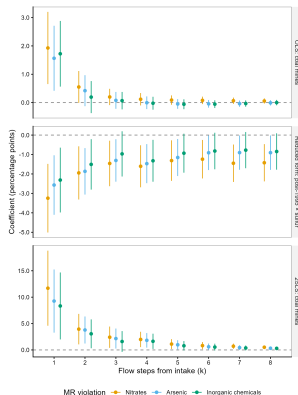
Enforcement actions show no consistent pattern for utilities upstream of a mine at any choice of k , supporting the exclusion restriction.

Appendix: concentration effects across choice of upstream watershed radius



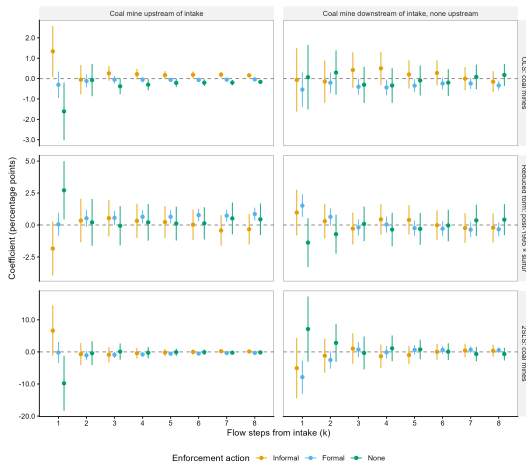
Arsenic and barium rise significantly only at $k = 1$ and are indistinguishable from zero at larger radii. Nitrate is insignificant at $k = 1$, significant from $k = 2$ to $k = 4$ (peaking at $k = 3$), and fades beyond; selenium is significant only at $k = 1$ and $k = 2$.

Appendix: MR violation effect across choice of upstream watershed radius



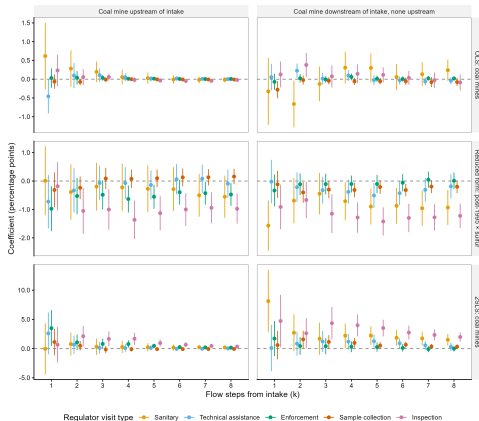
The 2SLS effect on MR violations is largest at $k = 1$ and declines smoothly as the upstream radius widens. Nitrate stays positive and significant at every k ; arsenic and inorganic chemical estimates stay positive but are borderline or insignificant at some radii of $k \geq 3$.

Appendix: enforcement effect across choice of upstream watershed radius



The formal-enforcement estimate is not stable in sign or significance across choices of k , consistent with the fixed-effects sensitivity found in the main specification.

Appendix: visit effect across choice of upstream watershed radius



The rise in inspection visits downstream of mining is significant at every $k \geq 2$ but not at $k = 1$. Sanitary visits remain insignificant throughout, while enforcement visits are significant at $k = 1$ and at most radii beyond $k = 2$.

Appendix: who appears in the Six-Year Review

Panel A: 1997–2005

	Utility appears in SYR2		Diff.	<i>p</i> -value
	Yes (1)	No (2)		
Ever MR violation	32.2%	18.0%	14.2%***	0.000
Ever MCL violation	1.7%	1.2%	0.5%	0.619
Mean annual violations (count, 0–6)	0.09	0.09	-0.00	0.874

Panel B: 1985–2005

	Utility appears in SYR2		Diff.	<i>p</i> -value
	Yes (1)	No (2)		
Ever MR violation	51.3%	44.1%	7.2%*	0.090
Ever MCL violation	5.5%	3.6%	2.0%	0.277
Mean annual violations (count, 0–6)	0.13	0.20	-0.07**	0.011

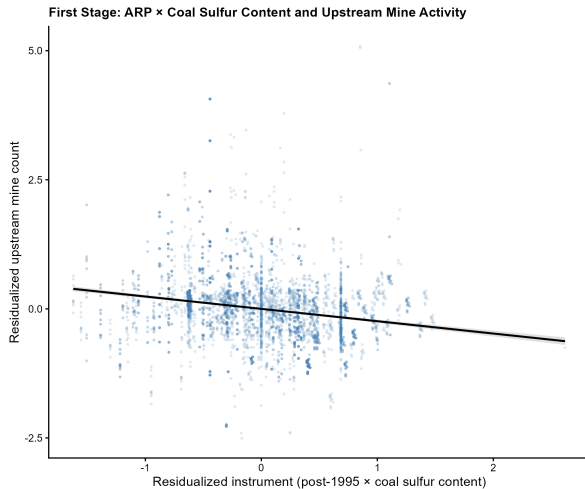
Utilities with SYR2 readings are 7–14 pp more likely to ever incur an MR violation. The concentration sample over-represents the more negligent utilities, so it is not selected toward clean systems.

Appendix: enforcement and visits in the panel

	Whole panel		During MR year		During MCL year	
	%	N	%	N	%	N
Panel A: Enforcement type						
Formal	3.00	323	9.41	70	6.12	3
Informal	18.24	1,965	66.67	496	59.18	29
Any	20.75	2,235	70.70	526	67.35	33
Panel B: Visit type						
Sanitary	13.59	1,464	13.17	98	10.20	5
Technical assistance	2.29	247	4.70	35	2.04	1
Enforcement	1.05	113	2.02	15	4.08	2
Sample collection	0.78	84	0.54	4	0.00	0
Inspection	3.43	369	5.65	42	2.04	1

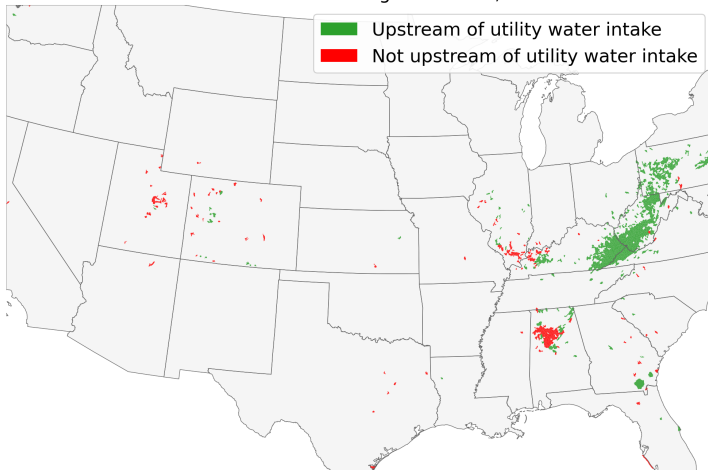
Enforcement responds to violations — any enforcement rises from 21% overall to 71% in MR years — but sanitary visits are no more likely in MR years than in any other year.

Appendix: residualized first stage



Appendix: sample watersheds

HUC12 watersheds containing coal mines, 1985-2005



Appendix: Corollary 1, statement and proof

Write the share of types under-reporting as $h(\theta, m) = 1 - G(c^*(\theta); m)$, with cutoff $c^*(\theta) = t - (r - ps)q(a_{SR}(\theta))$, so that

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m) d\theta}_{\text{A: cost}} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m} d\theta}_{\text{B: composition}}.$$

Corollary 1. If $q(a_{SR}(\theta)) \equiv \bar{q}$ on $(0, \bar{\theta})$, then $\partial MR / \partial m = -\partial G(\bar{c}; m) / \partial m \geq 0$, where $\bar{c} = t - (r - ps)\bar{q}$.

Proof. The cutoff is then type-invariant, $c^*(\theta) \equiv \bar{c}$, so $h(\theta, m) = 1 - G(\bar{c}; m)$ does not vary with θ and comes outside both integrals. The limits 0 and $\bar{\theta}$ do not depend on m , so by Leibniz's rule

$$B = [1 - G(\bar{c}; m)] \frac{d}{dm} \left[\int_0^{\bar{\theta}} f(\theta; m) d\theta \right] = [1 - G(\bar{c}; m)] \frac{d}{dm} [1] = 0,$$

since $f(\cdot; m)$ is a density for every m . Channel A collapses to $-\partial G(\bar{c}; m) / \partial m \geq 0$, because contamination shifts reporting costs up. $\square$

Why it matters. Channel B is the concealment margin: it operates only if contamination reallocates mass toward types whose exceedance risk, and hence whose cutoff, differs. Shut it down and any remaining rise in under-reporting is avoidance of the reporting technology itself. The 50%-of-MCL nitrate step is the empirical counterpart — monitoring frequency jumps, the sanction schedule does not, and readings are nowhere near the limit.

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